<u>25CS1302E</u> - DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

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<u>LAB COURSEWORK</u>

<u>Q.</u> **Week 2:**

The Librarian’s Dashboard Scenario The library is now digital! The head librarianneeds a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new bookis added and assigned to a category at the exact same time—if one part fails, thewhole process must stop to keep the catalog accurate.

Task to be Done

1. TheCatalog Search (JOINs): Create a third table called Loans that connects member_idto book_id. Write
a query to show the Member Name and the Book Title they havecurrently borrowed.
2. Collection Stats (Aggregations): Write a query to find thetotal number of books published in each year. o
Hint: Use COUNT(book_id) andGROUP BY published_year.
3. The Donation Transaction (ACID): Use aTransaction (BEGIN...COMMIT) to add a new book to the
Books table andsimultaneously log the entry in a Donation_History table.
4. Search Speed(Indexing): Create an Index on the isbn column. Explain why searching by ISBN isnow
faster for the librarian than searching by Title. Skill Set Gained • RelationalLogic: Connecting independent
tables (Books + Members) through a Junction table(Loans). • Analytical SQL: Using GROUP BY to generate
business insights. •Atomic Operations: Using Transactions to ensure multiple database changessucceed or
fail as a single unit.

SQL: mysql> USE bookflow_db;
Database changed
mysql>
mysql> SHOW TABLES;

```
+-----------------------+
| Tables_in_bookflow_db |
+-----------------------+
| books                 |
| members               |
+-----------------------+
2 rows in set (0.005 sec)
mysql> SELECT * FROM books;
+---------+---------------+---------------+----------------+
| book_id | title         | isbn          | published_year |
+---------+---------------+---------------+----------------+
|       1 | The Alchemist | 9780061122415 |           1988 |
|       2 | Clean Code    | 9780132350884 |           2008 |
|       3 | Atomic Habits | 9780735211292 |           2018 |
+---------+---------------+---------------+----------------+
3 rows in set (0.001 sec)
mysql> SELECT * FROM members;
+-----------+--------------+--------------------------+
| member_id | full_name    | email                    |
+-----------+--------------+--------------------------+
|         1 | Anil Kumar   | anil.kumar@example.com   |
|         2 | Priya Sharma | priya.sharma@example.com |
|         3 | Ravi Verma   | ravi.verma@example.com   |
+-----------+--------------+--------------------------+
3 rows in set (0.000 sec)
mysql> CREATE TABLE Loans (
    ->     loan_id INT PRIMARY KEY,
    ->     member_id INT,
    ->     book_id INT,
    ->     loan_date DATE,
    ->     FOREIGN KEY (member_id) REFERENCES members(member_id),
    ->     FOREIGN KEY (book_id) REFERENCES books(book_id)
    -> );
Query OK, 0 rows affected (0.027 sec)
mysql> SHOW TABLES;
+-----------------------+
| Tables_in_bookflow_db |
+-----------------------+
| books                 |
| Loans                 |
```

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| members              |
+----------------------+
3 rows in set (0.001 sec)
mysql> DESCRIBE Loans;
+-----------+------+------+-----+---------+-------+
| Field     | Type | Null | Key | Default | Extra |
+-----------+------+------+-----+---------+-------+
| loan_id   | int  | NO   | PRI | NULL    |       |
| member_id | int  | YES  | MUL | NULL    |       |
| book_id   | int  | YES  | MUL | NULL    |       |
| loan_date | date | YES  |     | NULL    |       |
+-----------+------+------+-----+---------+-------+
4 rows in set (0.006 sec)
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
    -> VALUES
    -> (1, 1, 1, '2025-01-05'),
    -> (2, 2, 2, '2025-01-08'),
    -> (3, 3, 3, '2025-01-10'),
    -> (4, 1, 2, '2025-02-01'),
    -> (5, 2, 1, '2025-02-05'),
    -> (6, 3, 2, '2025-02-12'),
    -> (7, 1, 3, '2025-03-01'),
    -> (8, 2, 3, '2025-03-07'),
    -> (9, 3, 1, '2025-03-15'),
    -> (10, 1, 1, '2025-04-01');
Query OK, 10 rows affected (0.007 sec)
Records: 10  Duplicates: 0  Warnings: 0
mysql> SELECT * FROM Loans;
+---------+-----------+---------+------------+
| loan_id | member_id | book_id | loan_date  |
+---------+-----------+---------+------------+
|       1 |         1 |       1 | 2025-01-05 |
|       2 |         2 |       2 | 2025-01-08 |
|       3 |         3 |       3 | 2025-01-10 |
|       4 |         1 |       2 | 2025-02-01 |
|       5 |         2 |       1 | 2025-02-05 |
|       6 |         3 |       2 | 2025-02-12 |
|       7 |         1 |       3 | 2025-03-01 |
|       8 |         2 |       3 | 2025-03-07 |
|       9 |         3 |       1 | 2025-03-15 |
|      10 |         1 |       1 | 2025-04-01 |
+---------+-----------+---------+------------+
10 rows in set (0.001 sec)
mysql> SELECT m.full_name AS Member_Name,
```

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    ->        b.title AS Book_Title
    -> FROM Loans l
    -> INNER JOIN members m
    -> ON l.member_id = m.member_id
    -> INNER JOIN books b
    -> ON l.book_id = b.book_id;
+--------------+---------------+
| Member_Name  | Book_Title    |
+--------------+---------------+
| Ravi Verma   | The Alchemist |
| Priya Sharma | The Alchemist |
| Anil Kumar   | The Alchemist |
| Anil Kumar   | The Alchemist |
| Ravi Verma   | Clean Code    |
| Priya Sharma | Clean Code    |
| Anil Kumar   | Clean Code    |
| Ravi Verma   | Atomic Habits |
| Priya Sharma | Atomic Habits |
| Anil Kumar   | Atomic Habits |
+--------------+---------------+
10 rows in set (0.001 sec)
mysql> SELECT published_year,
    ->        COUNT(book_id) AS Total_Books
    -> FROM books
    -> GROUP BY published_year
    -> ORDER BY published_year;
+----------------+-------------+
| published_year | Total_Books |
+----------------+-------------+
|           1988 |           1 |
|           2008 |           1 |
|           2018 |           1 |
+----------------+-------------+
3 rows in set (0.003 sec)
mysql> CREATE TABLE Donation_History (
    ->     donation_id INT PRIMARY KEY,
    ->     book_id INT,
    ->     donor_name VARCHAR(100),
    ->     donation_date DATE,
    ->     FOREIGN KEY (book_id) REFERENCES books(book_id)
    -> );
Query OK, 0 rows affected (0.011 sec)
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> INSERT INTO books (title, isbn, published_year)
    -> VALUES ('Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.003 sec)
mysql> SELECT book_id
    -> FROM books
    -> WHERE isbn = '9780451526342';
+---------+
| book_id |
+---------+
|       5 |
+---------+
1 row in set (0.001 sec)
mysql> INSERT INTO Donation_History
    -> (donation_id, book_id, donor_name, donation_date)
    -> VALUES
    -> (1, 5, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.004 sec)
mysql> COMMIT;
Query OK, 0 rows affected (0.002 sec)
mysql> SELECT * FROM books;
+---------+---------------+---------------+----------------+
| book_id | title         | isbn          | published_year |
+---------+---------------+---------------+----------------+
|       1 | The Alchemist | 9780061122415 |           1988 |
|       2 | Clean Code    | 9780132350884 |           2008 |
|       3 | Atomic Habits | 9780735211292 |           2018 |
|       5 | Animal Farm   | 9780451526342 |           1945 |
+---------+---------------+---------------+----------------+
4 rows in set (0.000 sec)
mysql> SELECT * FROM Donation_History;
+-------------+---------+------------+---------------+
| donation_id | book_id | donor_name | donation_date |
+-------------+---------+------------+---------------+
|           1 |       5 | Raj Kumar  | 2026-07-31    |
+-------------+---------+------------+---------------+
1 row in set (0.000 sec)
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)
mysql> INSERT INTO books (title, isbn, published_year)
    -> VALUES ('Temporary Book', '1111111111111', 2025);
Query OK, 1 row affected (0.001 sec)
mysql> SELECT * FROM books
    -> WHERE title = 'Temporary Book';
+---------+----------------+---------------+----------------+
```

```
| book_id | title          | isbn          | published_year |
+---------+----------------+---------------+----------------+
|       6 | Temporary Book | 1111111111111 |           2025 |
+---------+----------------+---------------+----------------+
1 row in set (0.000 sec)
mysql> ROLLBACK;
Query OK, 0 rows affected (0.001 sec)
mysql> SELECT * FROM books
    -> WHERE title = 'Temporary Book';
Empty set (0.000 sec)
mysql> CREATE INDEX idx_title
    -> ON books(title);
Query OK, 0 rows affected (0.029 sec)
Records: 0  Duplicates: 0  Warnings: 0
mysql> SHOW INDEX FROM books;
+-------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+---
---------+---------+---------------+---------+------------+
| Table | Non_unique | Key_name  | Seq_in_index | Column_name | Collation | Cardinality |
Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+---
---------+---------+---------------+---------+------------+
| books |          0 | PRIMARY   |            1 | book_id     | A         |           5 |     NULL |   NULL |      |
BTREE      |         |               | YES     | NULL       |
| books |          0 | isbn      |            1 | isbn        | A         |           5 |     NULL |   NULL |      | BTREE
|         |               | YES     | NULL       |
| books |          1 | idx_title |            1 | title       | A         |           4 |     NULL |   NULL |      | BTREE
|         |               | YES     | NULL       |
+-------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+---
---------+---------+---------------+---------+------------+
3 rows in set (0.005 sec)
mysql>
```